

Artificial Intelligence: A Brief Overview

What is Artificial Intelligence?

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines. These machines are programmed to think and learn like humans, performing tasks that typically require human cognition such as visual perception, speech recognition, decision-making, and language translation.

Types of AI

1. Narrow AI: Designed for specific tasks like facial recognition or chess. Examples include Siri, Alexa, and recommendation systems used by Netflix and Spotify.
2. General AI: Hypothetical AI that can perform any intellectual task a human can. This does not yet exist but is a major research goal.
3. Super AI: A theoretical form of AI that surpasses human intelligence in all areas. This remains speculative.

Machine Learning

Machine Learning (ML) is a subset of AI that enables machines to learn from data without being explicitly programmed. Key techniques include supervised learning, unsupervised learning, and reinforcement learning. Popular frameworks include TensorFlow, PyTorch, and scikit-learn.

Large Language Models

Large Language Models (LLMs) like GPT-4 and Claude are trained on massive text datasets and can generate human-like text, answer questions, write code, and perform complex reasoning tasks. They use transformer architecture introduced by Google in 2017.